Beyond the Streetlight

by Byrne, Corrado, Kennickell

REMARKs by Christopher D. Carroll

Formerly of the Division of Research and Statistics: 1990-95

Johns Hopkins University (1995-2023)

Federal Reserve RS100 Conference

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https://econ-ark.org/materials/beyond-the-streetlight

Thank you to all Federal Reserve Staff

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- For doing a vitally important job as well as humanly possible

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- It's data all the way!

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I would add a Tertiary:

- books $\in \{$Green, Teal, Beige$\}$
- staff memos ...

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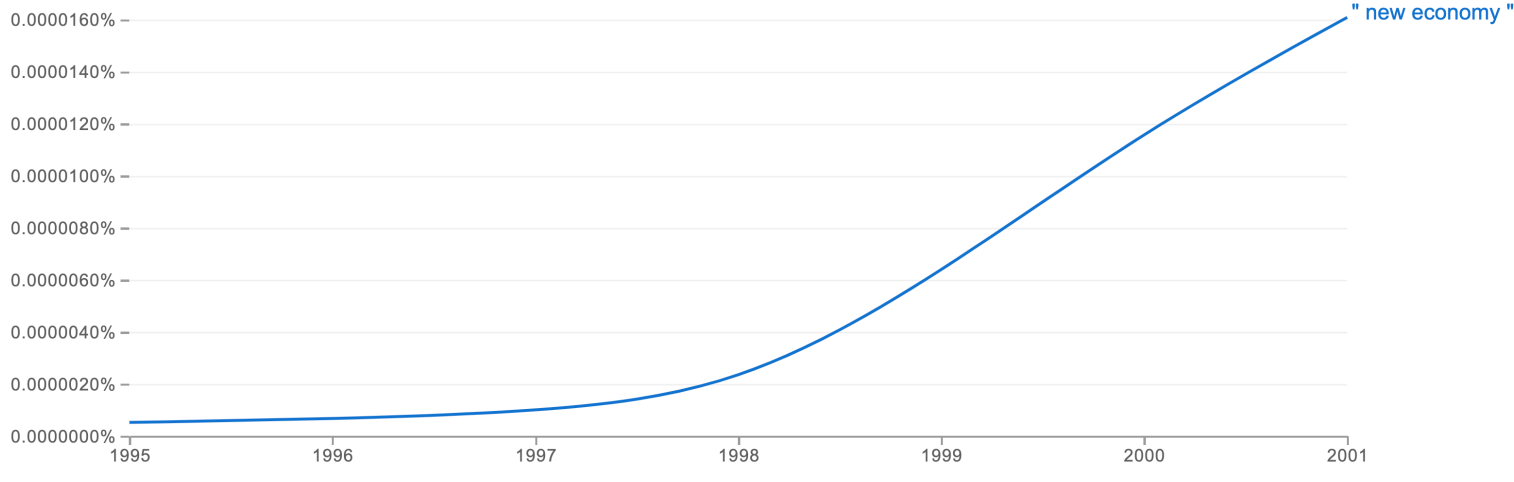
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- 'opportunistic' disinflation

Google nGram of 'new economy'



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- Or has the Fed gotten better over time?
    - That's one point of better measurement
- Maybe everybody has gotten better over time?
    - Measurement improvements make everybody better off

The Data: Forecasts and What Happened

Forecasts made each quarter, 1982Q1 to 2017Q4 (144 quarters), from the Philadelphia Fed

- **Greenbook/Tealbook (GB)**: the Fed staff's forecast, from the later of each quarter's two Greenbooks, which is more likely to come after the SPF's survey (Greenbooks become public after five years)
- **Survey of Professional Forecasters (SPF)**: the mean forecast
- Each forecasts real PCE growth (annualized) for this quarter and the next three, and the unemployment rate for this quarter and four quarters ahead

What happened, from FRED, retrieved 2023-11-03

- `UNRATE`: the unemployment rate, averaged over each quarter
- `DPCERA3Q086SBEA`: real PCE, as annualized quarterly growth

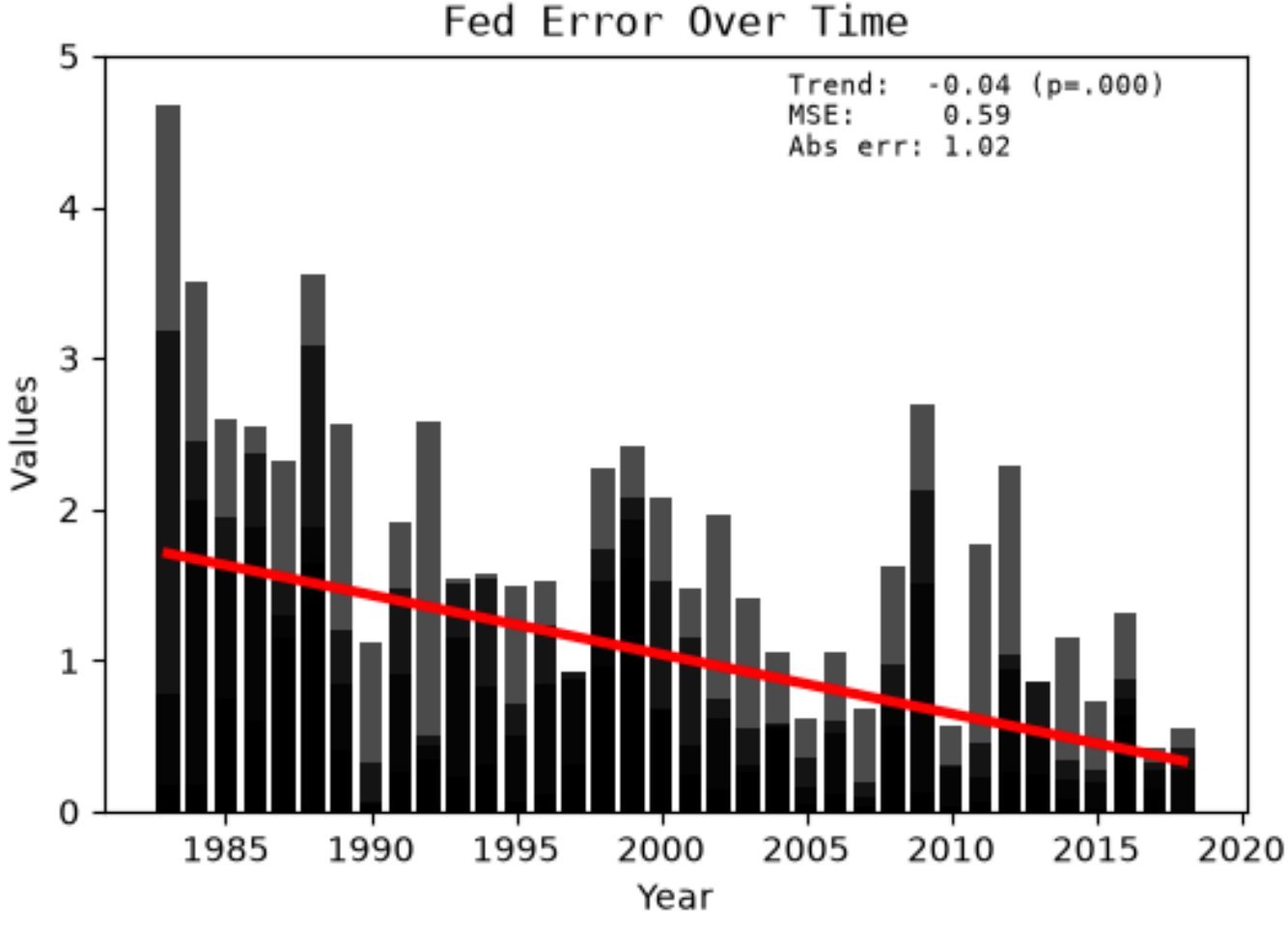
From Forecasts to Errors, in pandas

Step	In the code
Read a forecaster's spreadsheet, one sheet per variable	`pd.ExcelFile(path).parse("gRPCE")`
Keep the later Greenbook in each quarter	`results_df.iloc[1::2]`
Year-ahead forecast: the mean of the four quarterly forecasts	`GB_cons.mean(axis=1)`
What happened: growth averaged over the same four quarters	`FRED[col].rolling(window=4).mean()`
Unemployment: the forecast and the actual four-quarter change	`UNEMPF4 - UNEMPF0` and `FRED[col].diff(periods=4)`
The absolute error	`np.abs(forecast - actual)`
Its trend: `X` is a constant and the year minus its mean	`sm.OLS(error, X).fit(cov_type="HC3")`

The scripts are in `code/main/`. They line the series up by row position, after `reset_index(drop=True)`, and date each forecast four quarters after it was made, so the forecasts made in 1982Q1–2017Q4 are plotted as 1983–2018.

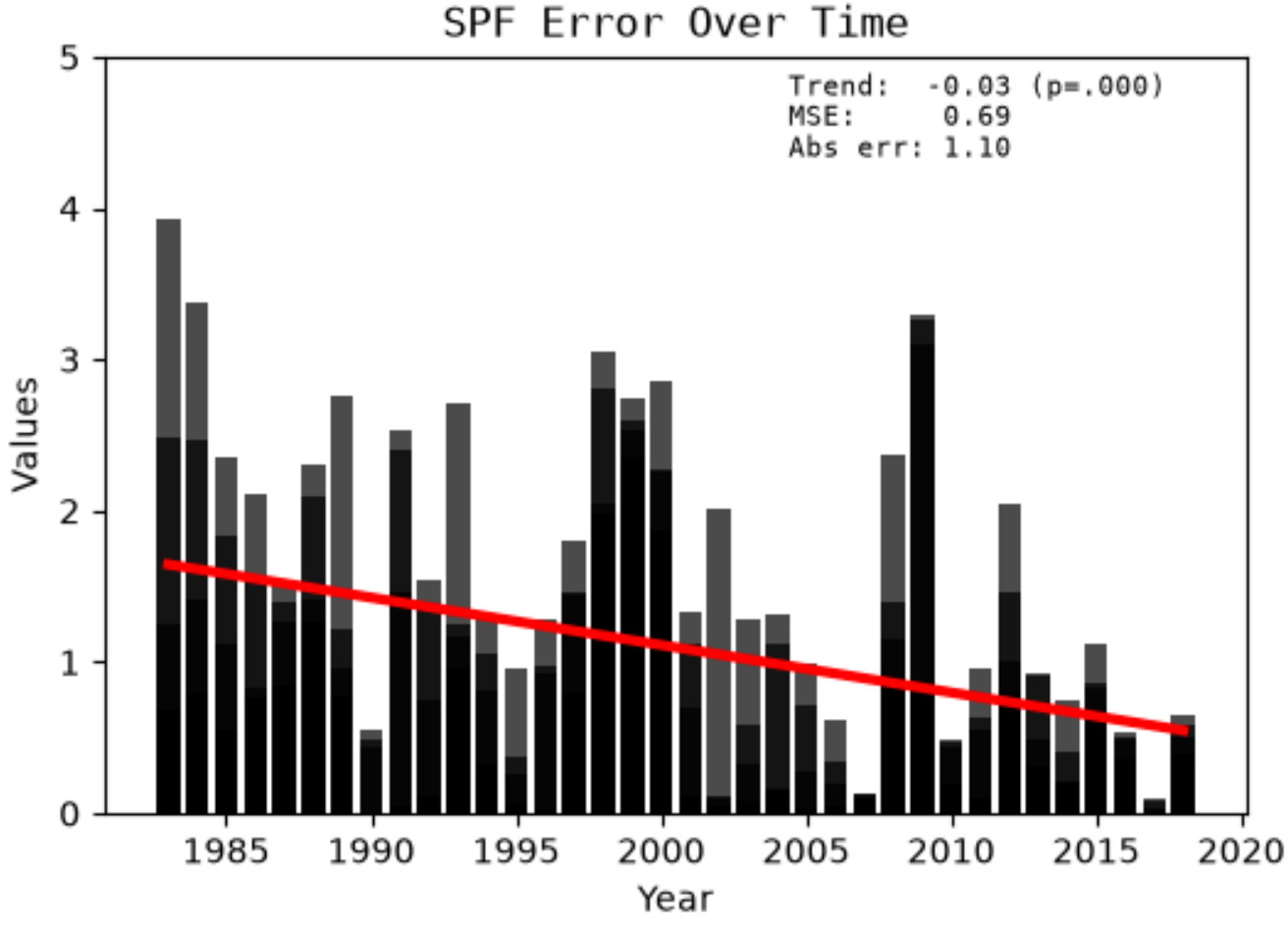
Tertiary Data from Philly Fed and FRED

Forecasts from books ∈ {Green, Teal}



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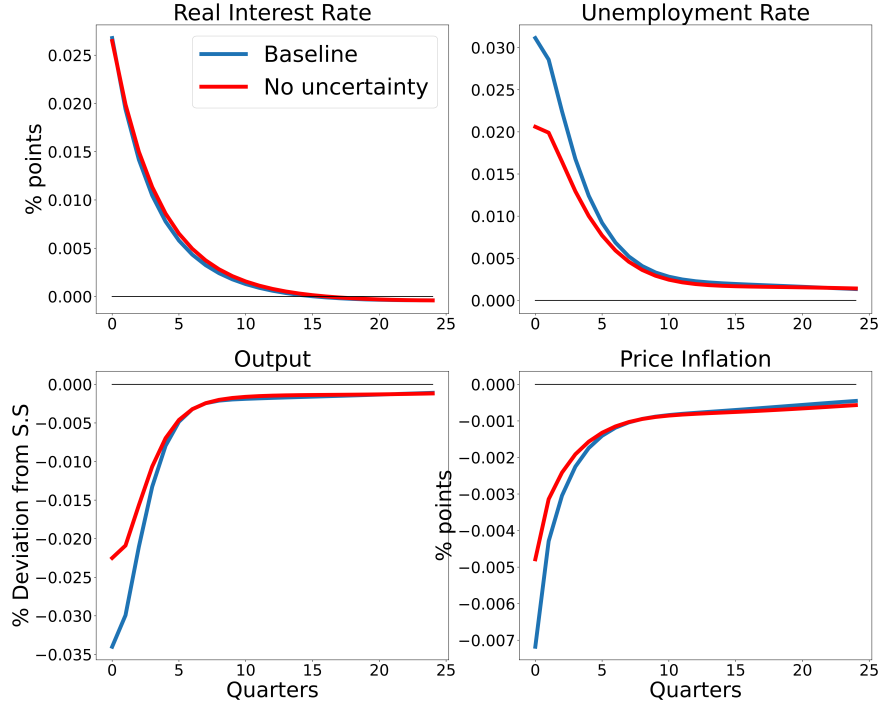
Forecasts from SPF



How Much of Business Cycle is Driven by Uncertainty?

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2.5 bp increase in FF in Will Du's 'HANK and SAM' model.



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- Utility cost of business cycles directly proportional to c
    - $\Delta V \approx u'(c) \times \Delta c$

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- Utility cost of business cycles directly proportional to c
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- Better measurement $\Rightarrow$ less uncertainty $\Rightarrow$ higher welfare

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- Better *micro* data
    - linked to macro through RDC